

Sara Tahajomi Banafshehvaragh

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[Google Scholar](#)

RESEARCH PROFILE

Ph.D. Candidate in Civil and Environmental Engineering at Old Dominion University. Research focuses on landslide hazard assessment, geotechnical earthquake engineering, ground-motion analysis, and the application of machine learning and geospatial methods to natural-hazard evaluation and infrastructure risk assessment.

EDUCATION

Ph.D. in Civil and Environmental Engineering

Old Dominion University, Norfolk, VA, USA, Graduate Research Assistant / Teaching Assistant

Start Date: January 2023

Expected Graduation: 2028

Research Focus: Seismic Hazard Assessment, Ground Motion Processing, Geotechnical Engineering, Landslide Hazard Assessment, Machine Learning Applications in Civil Engineering.

Relevant Coursework:

Geotechnical Earthquake Engineering, Soil Dynamics, Advanced Soil Mechanics, Slope Stability and Earth Retaining Structures, Foundation Engineering, Statistical Processing, Regression and Time Series, Statistical/Probability Models for Data Science and Analytics

Master of Science in Geotechnical Engineering

University of Science and Culture

Sep 2019- Sep 2021

Thesis title: Investigation of the Effects of Land Subsidence on the Safety of Electric Transmission Structures

Bachelor of Science in Civil Engineering

University of Science and Culture

Sep 2014- Sep 2018

RESEARCH INTERESTS

Geotechnical Earthquake Engineering; Seismic Hazard Evaluation and Risk Assessment; Ground Motion Analysis and Classification; Landslide Hazard Assessment; Probabilistic Hazard Modeling; Machine Learning for Natural Hazard Assessment; Infrastructure Resilience and Risk Mitigation.

PUBLICATIONS

Journal Articles:

Akbari Garakani, A., Tahajomi Banafshehvaragh, S., Saheb, S., Sadeghi, H. 2025. Assessing the Vulnerability of Power Transmission Towers to Land Subsidence and Forecasting Future Trends Using Multi-Source Datasets: Insights from Moein Abad, Iran. In *Innovative Infrastructure Solutions*, Springer.

Conference Proceedings:

Tahajomi Banafshehvaragh, S., & Wang, P. (2026). A Novel Machine Learning Approach for Automated Usability Classification of Ground Motion Recordings. *Forthcoming Proceedings of the 13th National Conference on Earthquake Engineering (13NCEE)*, Earthquake Engineering Research Institute, Portland, OR. (Accepted for publication).

Tahajomi Banafshehvaragh, S., Wang, P. 2025. A New Probabilistic Based Approach for Rainfall-triggered Landslide Hazard Assessment. In *Geotechnical Frontiers 2025*.

Tahajomi Banafshehvaragh, S., Wang, P. 2024. Statewide Seismic-Induced Landslide Risk Assessment in Virginia: A Probabilistic Approach. In *Graduate Research Achievement Day 2024*, Old Dominion University.

Akbari Garakani, A., Tahajomi Banafshehvaragh, S., Yeganeh Fallah, A. 2021. The Imposed Costs of Land Subsidence on High-Voltage Transmission Lines. In *Geotechnics Fundamentals and Applications in Construction: New Materials, Structures, Technologies and Calculations*.

RESEARCH EXPERIENCE

Machine Learning Framework for Automated Ground-Motion Classification

- Developed a machine-learning framework for automated usability classification of earthquake ground-motion records to support seismic hazard and ground-motion analysis.
- Engineered signal-processing and pattern-recognition features that capture waveform quality, phase completeness, and seismic signal characteristics across approximately 9,000 global recordings.
- Built automated Python-based workflows for large-scale ground-motion processing, feature extraction, model development, and validation.
- Achieved classification accuracy comparable to expert manual identification by trained analysts.

Probabilistic-Based Rainfall-Triggered Landslide Hazard Assessment

- Developed a probabilistic approach for evaluating rainfall-triggered landslide hazards by combining susceptibility modeling with rainfall-threshold analysis.
- Integrated topographic, geological, land-use, and precipitation datasets within a GIS framework to assess landslide hazards across Virginia.
- Produced high-resolution statewide hazard maps to support infrastructure planning, hazard mitigation, and risk-informed decision-making.

HVSR-Based Site Characterization and Coastal Land Subsidence Research

- Conducted field and analytical investigations using microtremor horizontal-to-vertical spectral ratio (HVSR) techniques to support the development of non-intrusive site characterization methods in coastal environments.
- Collected and analyzed ambient vibration data at sites with co-located CPT measurements to investigate relationships between HVSR-derived parameters and subsurface conditions.
- Processed microtremor recordings to estimate fundamental site frequencies and site-response characteristics.

INDUSTRY EXPERIENCE

Structural Engineering Intern – Mott MacDonald (Arlington, VA | May 2026 – August 2026)

- Performed PLAXIS 2D/3D analyses for tunnel and underground infrastructure projects.
- Designed initial support systems for tunnel liner plate and steel casing installations.
- Interpreted boring logs, CPT data, laboratory test results, and geologic information to support geotechnical design.

Structural Engineering Intern – Mott MacDonald (Arlington, VA | June 2025 – August 2025)

- Performed PLAXIS analyses and supported tunnel, support-of-excavation, and geotechnical design for transportation and underground infrastructure projects.
- Contributed to geotechnical data review, ground model development, slope stability evaluations, GIS mapping, and construction submittal reviews.

TEACHING EXPERIENCE

Graduate Teaching Assistant – Soil Mechanics Lab (Spring 2023 – Fall 2024)

- Instructed undergraduate students in fundamental soil mechanics laboratory experiments, including Atterberg limits, grain size distribution, permeability tests, and direct shear strength analysis.
- Provided guidance on lab procedures, data collection, and report writing to enhance students' understanding of geotechnical engineering principles.

HONORS & AWARDS

- Ashraf's 13NCEE Participation Grant, Earthquake Engineering Research Institute (EERI), 2026
- Rollie Dubbé Engineering Scholarship, Old Dominion University, 2024–2025
- Full Tuition Waiver and Graduate Assistantship, Old Dominion University, 2023–Present
- Top 10% of Graduating Class, M.S. in Geotechnical Engineering, University of Science and Culture
- Top 2% in National University Entrance Examination, Iran

TECHNICAL SKILLS

Seismic & Hazard Analysis

Ground Motion Processing, SeismoSignal, DEEPSOIL, Probabilistic Hazard Modeling

Programming & Data Science

R, Python, Machine Learning, Statistical Modeling

Civil Engineering Software

QGIS, ArcGIS, GRASS GIS, PLAXIS 2D/3D, SAFE, ETABS, SAP2000, Civil 3D, AutoCAD, Google SketchUp

REFERENCES

Pengfei Wang

- Assistant Professor, Department of Civil and Environmental Engineering, Old Dominion University, Norfolk, VA, 23518, USA
- Email: p1wang@odu.edu

Yang Yang

- Geotechnical Engineer, HNTB, Parsippany, Philadelphia, PA, USA
- Former Postdoctoral Researcher, Temple University,
- Email: dienyoung@outlook.com